

SUMMARY

This video explains the mathematical concepts and instructional insights from Part 02 of the GATE 2026 Discrete Mathematics Crash Course. The session advances from foundational set theory toward relational structures, with emphasis on algebraic properties of sets and the formal treatment of multisets.

Key themes include:

Rigorous understanding of algebraic laws of sets.
Formal structure of multisets and multiplicity logic.
Operational definitions using max, min, and subtraction.
Transition roadmap toward relations and posets.

I. ADVANCED PROPERTIES OF SET OPERATIONS

1.1 Fundamental Laws

Idempotent Law

$$A \cup A = A$$

$$A \cap A = A$$

Identity Law

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Dominance Law

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

Complementation Rules

$$A \cap A' = \emptyset$$

$$A \cup A' = U$$

$$(A')' = A$$

1.2 Commutative, Associative, and Distributive Laws

Commutative Laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Set difference is not commutative.

Associative Laws

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Distributive Laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

1.3 Absorption and De Morgan's Laws

Absorption Law

$$A \cap (A \cup B) = A$$

$$A \cup (A \cap B) = A$$

De Morgan's Laws

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

II. THEORY AND APPLICATION OF MULTISSETS

2.1 Definition and Context

A multiset is an unordered collection where elements may appear multiple times. The number of occurrences of an element is called its multiplicity.

Application in DBMS

In SQL:

SELECT returns multisets (duplicates allowed).

SELECT DISTINCT returns sets (unique tuples).

2.2 Multiset Operations

Let M_i be multiplicity of element A_i in multiset P .

Let N_i be multiplicity of A_i in multiset Q .

Operation	Resulting Multiplicity of A_i	Logic
Union All	$\max(M_i, N_i)$	Maximum frequency preserved
Intersect All	$\min(M_i, N_i)$	Common minimum frequency
Minus All ($P - Q$)	$M_i - N_i$ if $M_i > N_i$ else 0	Subtraction, never negative

2.3 Size of a Multiset

Size = Sum of multiplicities.

Example

$$A = \{2 \cdot a, 1 \cdot b, 3 \cdot d\}$$

$$\text{Size} = 2 + 1 + 3 = 6$$

III. COMBINATORIAL CHALLENGES

3.1 Symmetric Difference Associativity

Students are encouraged to verify whether symmetric difference satisfies:

$$(A \Delta B) \Delta C = A \Delta (B \Delta C)$$

This can be validated using Venn diagrams.

3.2 Multiset Counting Problem

Given:

$$A = \{1, 2, 3, \dots, 10\}$$

Question:

Number of multisets of size 4.

Key distinction:

Order does not matter.

Repetition allowed.

This corresponds to combinations with repetition:

$$n+r-1 \text{ C } r$$

Here:

$$\begin{aligned} &10+4-1 \text{ C } 4 \\ &= 13 \text{ C } 4 \end{aligned}$$